TABLE NAME: EMPLOYEE

id	First_name	last_name	department	Salary	hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
4	Sarah	Brown	IT	58000	2021-03-25	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2015-02-14	Chicago
7	Robert	Wilson	Finance	59000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

Questions

1. SELECT Statement

Write a SQL query to retrieve all columns from the employee table

Solution:

```
SELECT *
FROM EMPLOYEE
```

Output

id	First_name	last_name	department	Salary	City
1	John	Doe	IT	55000	New York
2	Jane	Smith	HR	48000	Chicago
3	Mike	Johnson	Finance	60000	Los Angeles
4	Sarah	Brown	IT	58000	New York
5	David	White	Marketing	52000	San Francisco
6	Emily	Davis	IT	62000	Chicago
7	Robert	Wilson	Finance	89000	Houston
8	Jessica	Moore	HR	81000	Los Angeles
9	Daniel	Clark	Marketing	51000	Chicago
10	Laura	Hall	IT	80000	San Francisco

2 SELECT DISTINCT Statement

Write a SQL to find all the unique departments in the employee table

Solution:

```
SELECT DISTINCT Department
FROM Employee
```

Output:

Department
IT
HR
Finance
Marketing

3. ORDER BY Statement

Write a SQL query to retrieve all employees first and last names, ordered by salary in descending order

```
SELECT first_name,
       last_name
FROM EMPlayee
ORDER by Salary desc;
```

Output:

First_name	last_name
Emily	Davis
Mike	Johnson
Robert	Wilson
John	Doe
Sarah	Brown
Daniel	Clark
David	White
Jessica	Moore
Laura	Hall
Jane	Smith

4. LIMIT Statement

Write a SQL query to retrieve the top 5 highest-paid employee

Solution

```
SELECT *
FROM EMPlayee
order by salary desc
limit 5
```

Output

ID	First_name	last_name	Department	Salary	Hire_date	City
6	Emily	Davis	IT	62000	2015-02-14	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
7	Robert	Wilson	Finance	59000	2019-10-01	Moosbon
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2026-03-21	New York

5 WHERE Statement

Write a SQL query to find employee who work in the IT department

Solution:

```
SELECT *
FROM EMPlOYEE
WHERE department = 'IT'
```

Output

ID	First_name	last_name	Department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2026-03-21	New York
6	Emily	Davis	IT	62000	2015-02-14	Chicago
10	Laura	Hall	IT	50000	2020-08-10	San Francisco

6. AND Statement

Write a SQL query to find employee who work in the Finance department AND have a Salary greater than $8,000

Solution

```
SELECT *
FROM EMPLOYEE
WHERE department = 'Finance' and Salary > 58000
```

Output

ID	First_name	last_name	Department	Salary	Hire_date	City
3	Mike	Johnson	Finance	60000	2017-19-30	Los Angeles
7	Robert	Wilson	Finance	89000	2019-10-01	Houston

7. OR Statement

Write a SQL query to find employees who work in the HR department OR the marketing department

Solution

```
SELECT *
FROM EMPLOYEE
WHERE department = 'HR' OR department = 'Marketing'
```

Output:

ID	First_name	~~last_name~~ department	~~department~~ last_name	Salary	Hire_date	City
2	Jane	HR	Smith	48000	2019-07-20	Chicago
5	David	marketing	White	52000	2016-04-10	San Francisco
8	Jessica	HR	Moore	51000	2018-05-22	Los Angeles
9	Daniel	Marketing	Clark	53000	2022-06-01	Chicago

8. NOT Statement

Write a SQL query to find employees who do not work in the IT department

Solution

```
SELECT *
FROM EMPLOYEE
WHERE NOT department = 'IT'
```

Output

ID	First_name	last_name	Department	Salary	Hire_date	City
2	Jane	Smith	HR	48000	2019-07-20	Chicago
3	Mike	Johnson	Finance	60000	2017-09-30	Los Angeles
5	David	White	Marketing	52000	2016-04-10	San Francisco
7	Robert	Wilson	Finance	89000	2019-10-01	Houston
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago

9. IN Statement

Write a SQL query to find employees who are in the HR, IT or Finance departments

Solution

```
SELECT *
FROM Employee
WHERE department IN ('HR', 'IT', 'Marketing');
```

Output

ID	First_name	last_name	Department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
2	Jane	Smith	HR	48000	2019-07-20	Chicago
4	Sarah	Brown	IT	53000	2026-03-21	New York
5	David	White	Marketing	52000	2016-04-10	San Francisco
6	Emily	Davis	IT	62000	2018-02-14	Chicago
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
10	Laura	Hall	IT	60000	2020-08-10	San Francisco

10. Combining Conditions

Write a SQL query to find employees who are in the IT department, have a salary greater than 50 000 and are located in New York

Solution

```
SELECT *
FROM EMPLOYEE
WHERE Department = 'IT' and Salary > 50000 and City = 'New York'
```

Output

ID	First_name	last_name	Department	Salary	Hire_date	City
1	John	Doe	IT	55000	2018-06-15	New York
4	Sarah	Brown	IT	53000	2026-03-21	New York

11. Combining conditions WHERE, AND and ORDER BY

write a SQL query to retrieve the first and last names of employees who work in finance or marketing department, earn more than $2000, and order the results by salary in descending order

Solution

```
SELECT first_name,
       last_name
FROM EMPLOYEE
WHERE department IN ('Finance','Marketing')
AND salary > $2000
ORDER BY salary DESC;
```

Output

First_name	last_name
Mike	Johnson
Robert	Wilson
Daniel	Clark

12. Combining SELECT DISTINCT, WHERE and IN

Write a SQL query to find all the unique cities where employees work, excluding those in the IT and HR departments

Solution

```
SELECT DISTINCT city
FROM Employee
WHERE department NOT IN ('IT','HR')
```

Output

City
Chicago
Los Angeles
San Francisco
Houston

13. Combining WHERE, NOT, AND and ORDER BY

Write a SQL query to retrieve employee who are NOT in the Finance department, have a salary greater than 50000 and order the results by hire date in ascending order

Solution

```
SELECT *
FROM EMPLOYEE
WHERE department <> 'Finance'
AND salary > 50000
ORDER BY hire_date ASC;
```

Output

ID	First_name	last_name	Department	Salary	hire_date	City
6	Emily	Davis	IT	62000	2015-02-16	Chicago
5	David	White	Marketing	52000	2016-04-10	San Francisco
8	Jessica	Moore	HR	51000	2018-05-22	Los Angeles
1	John	Doe	IT	55000	2018-06-15	New York
9	Daniel	Clark	Marketing	53000	2022-06-01	Chicago
4	Sarah	Brown	IT	53000	2026-05-21	New York

14. Combine WHERE, OR, IN and LIMIT

Write a SQL query to find the first 3 employee who work in either Chicago or Los Angeles and belongs to the IT or marketing department

Solution

```
SELECT *
FROM EMPLOYEE
WHERE city IN ('chicago', 'Los Angeles')
AND department IN ('IT', 'Marketing')
LIMIT 3;
```

Outpute

ID	First_name	last_name	department	salary	Hire_date	city
6	Emily	Davis	IT	82000	2015-02-14	Chicago
9	Daniel	Clark	Marketing	82000	2022-06-01	Chicago